



DATS 6303 - Deep Learning



DRIVER DROWSINESS DETECTION

Team Members:

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OBJECTIVE

- A critical factor in many devastating **traffic accidents** is driver drowsiness, which leads to severe injuries, loss of life, and considerable financial burdens on families and the economy.
- To mitigate this critical safety issue, we developed a DL model trained to analyze real time visual cues from a driver's face, focusing on **eye closure** and **yawning** to detect and predict instances of drowsiness.

DATASET

- We used a merged dataset from various **Kaggle** sources, organized for detecting yawn and eye states. Thanks to **Abdelaziz Nebil's** for facilitating this resource.
- The dataset comprises approximately **70,000** labeled images, encompassing both **eye states** (**open** and **closed**) and **yawn states** (**yawning** and **not yawning**), distributed across training and testing sets.

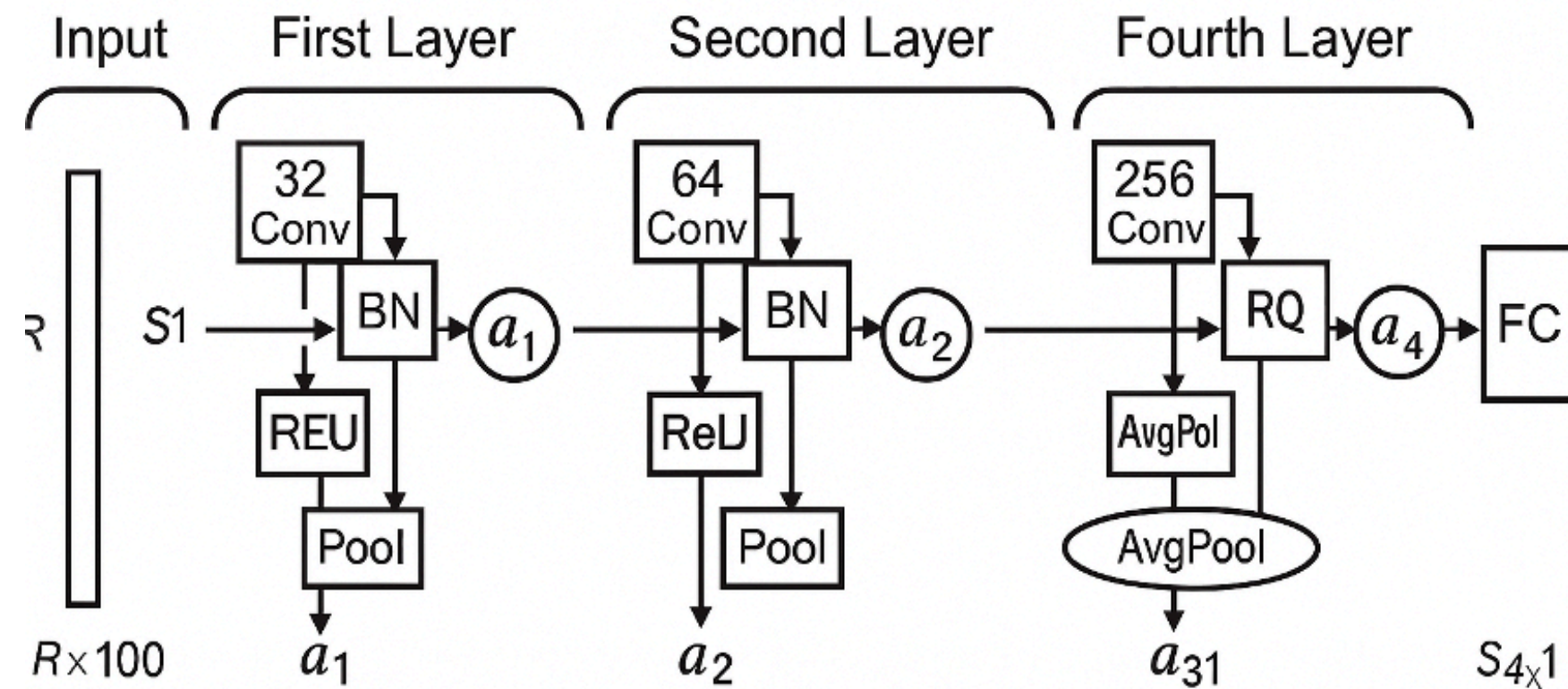
PROJECT GOALS

- To train a robust model for accurate **eye state** detection (**open/closed**).
- To train a separate model for reliable **yawn state** detection (**yawning/not yawning**).
- To implement a real-time system that **extracts** relevant **facial regions** from **live video** and utilizes the **trained models** for drowsiness prediction on each frame.

EYE STATE DETECTION

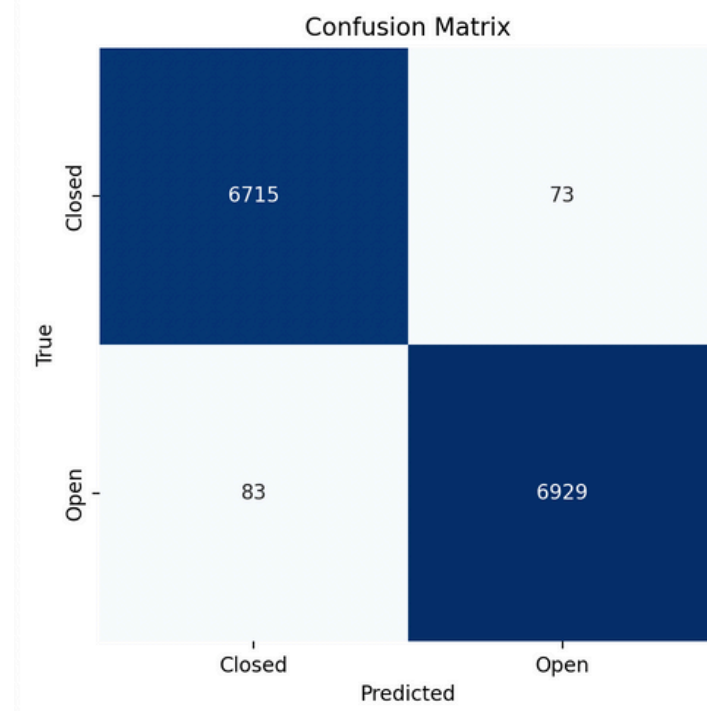
- We trained two models for **eye state detection** : a custom **CNN** and **ResNet18**, using augmented data.
- Our custom **CNN outperformed ResNet18** in terms of accuracy for eye state classification and learning.
- Conducted a analysis between the **CNN** and **ResNet18** models using **classification metrics** and **visual validation**
- We used **Grad-CAM** to visualize the specific eye regions the model focused on for its predictions.

EYE STATE - CNN



- Consists of **4 convolutional blocks** with **BatchNorm**, **ReLU**, and **Dropout**
- Final layer uses Adaptive Average Pooling followed by a fully connected layer

EYE STATE DETECTION COMPARISON - CNN VS RESNET18



```
Final Test Classification Report (Best Model Loaded):
      precision    recall  f1-score   support

   Closed      0.99      0.99      0.99      6788
    Open      0.99      0.99      0.99      7012

 accuracy      0.99      13800
```

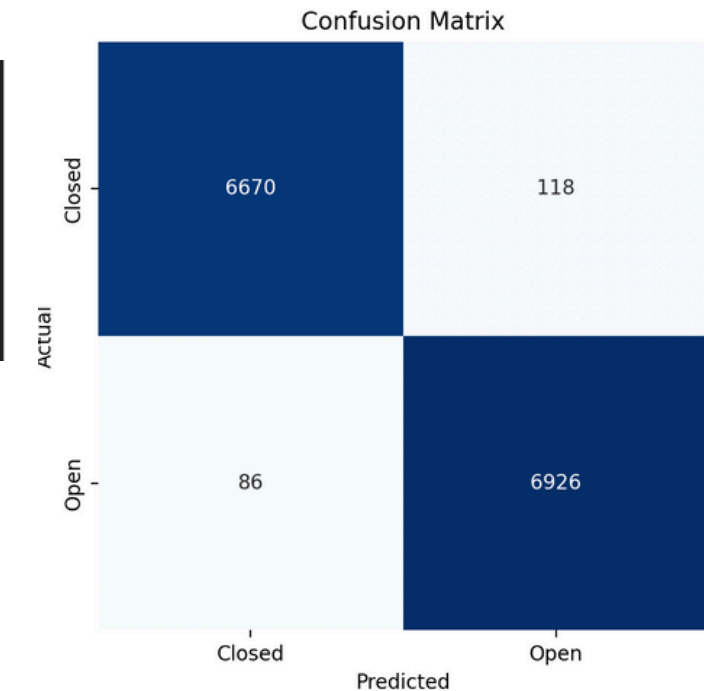
CNN

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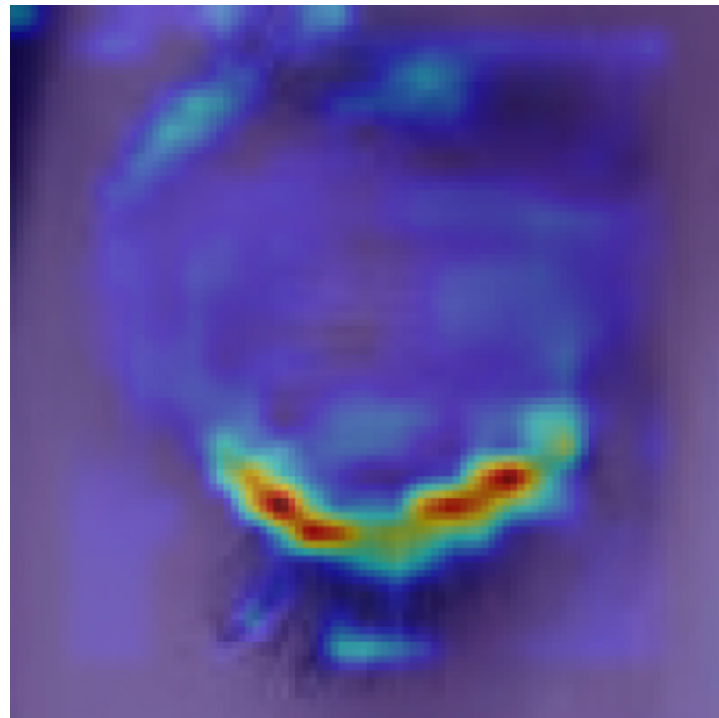
RESNET18



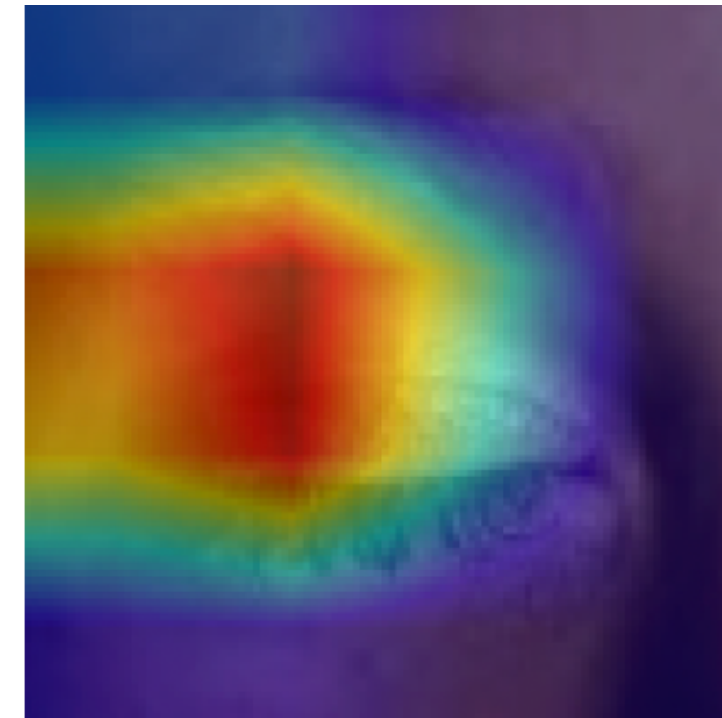
- **Both models** achieved an overall accuracy of **~99%** on the test set
- F1-score, precision, and recall are nearly identical for both models: **~0.99**
- No significant performance gap in terms of standard classification metrics
- Metrics alone suggest equivalent model performance in detecting open vs closed eyes

EYE STATE (CLOSED)

GRAD CAM - CNN VS RESNET18



CNN

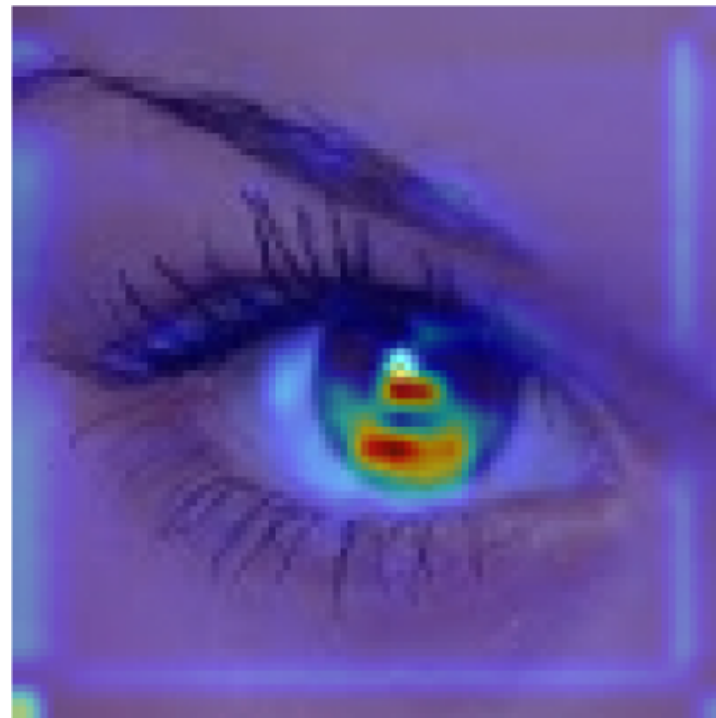


RESNET18

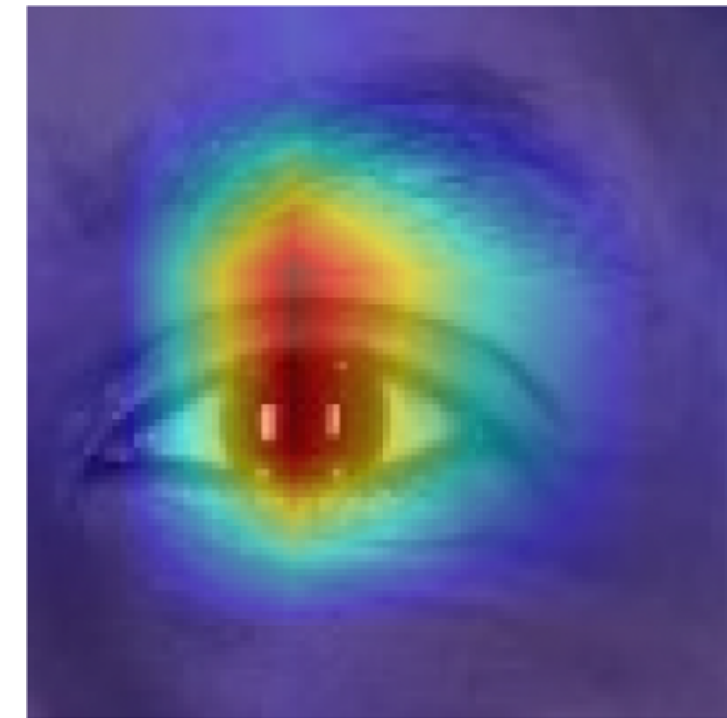
- Custom **CNN focuses sharply** on the **eye region**, showing **precise** localization
- **ResNet18** displays a **more diffused attention** map, sometimes **covering irrelevant areas**
- ResNet's pretrained filters may introduce generalized or noisy focus.
- Visual explainability shows that **CNN** learns more **interpretable, eye-centric features**

EYE STATE (OPEN)

GRAD CAM - CNN VS RESNET18



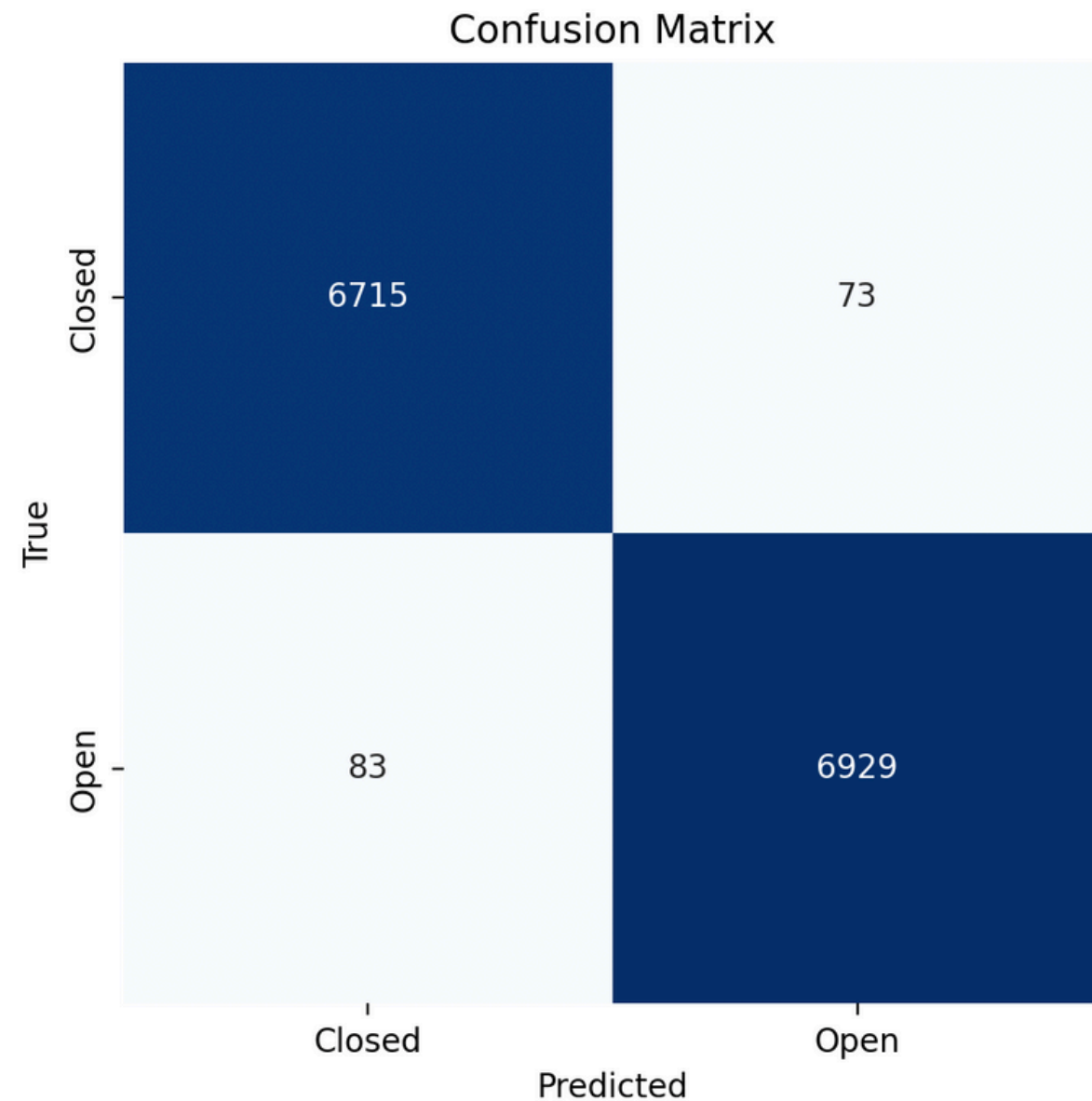
CNN



RESNET18

- **CNN** shows **tight** heatmap **focus** on **iris** and **eyelid**, **aligning** with **task** objective
- **ResNet18**'s attention is **wider** and **more scattered**, covering **upper face** or **eyebrow**
- Clear indication that **CNN** is **learning class-specific**, discriminative regions
- Supports the trend seen in closed-eye images: CNN is more visually consistent

CONFUSION MATRIX - CNN

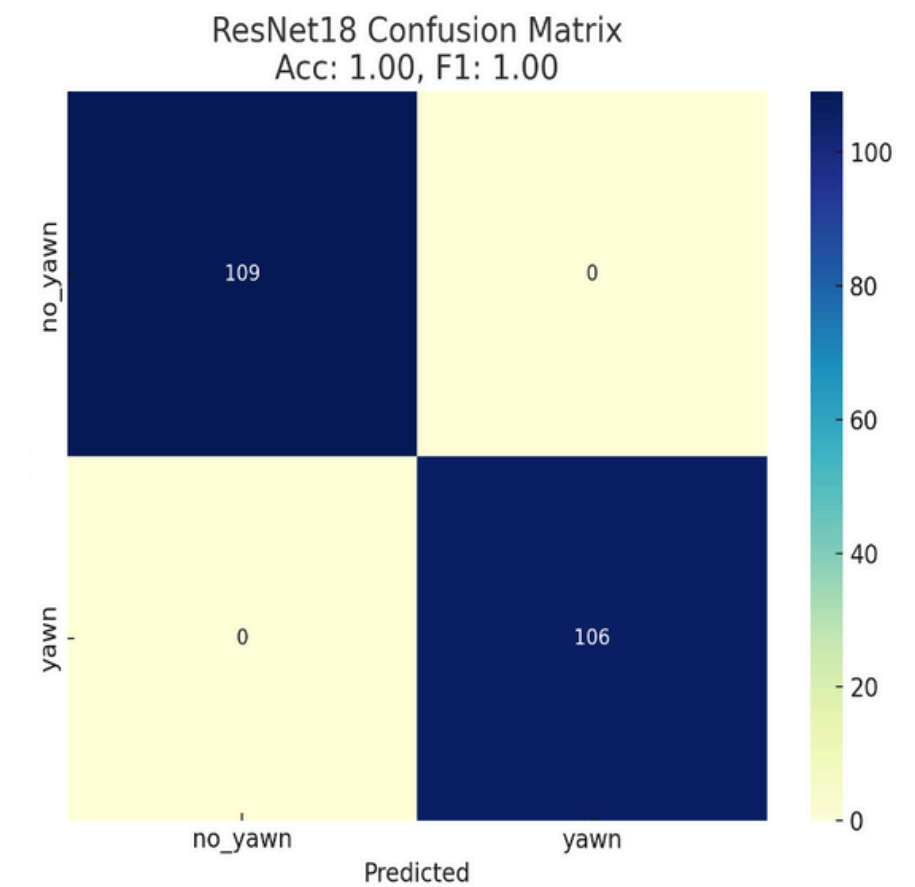
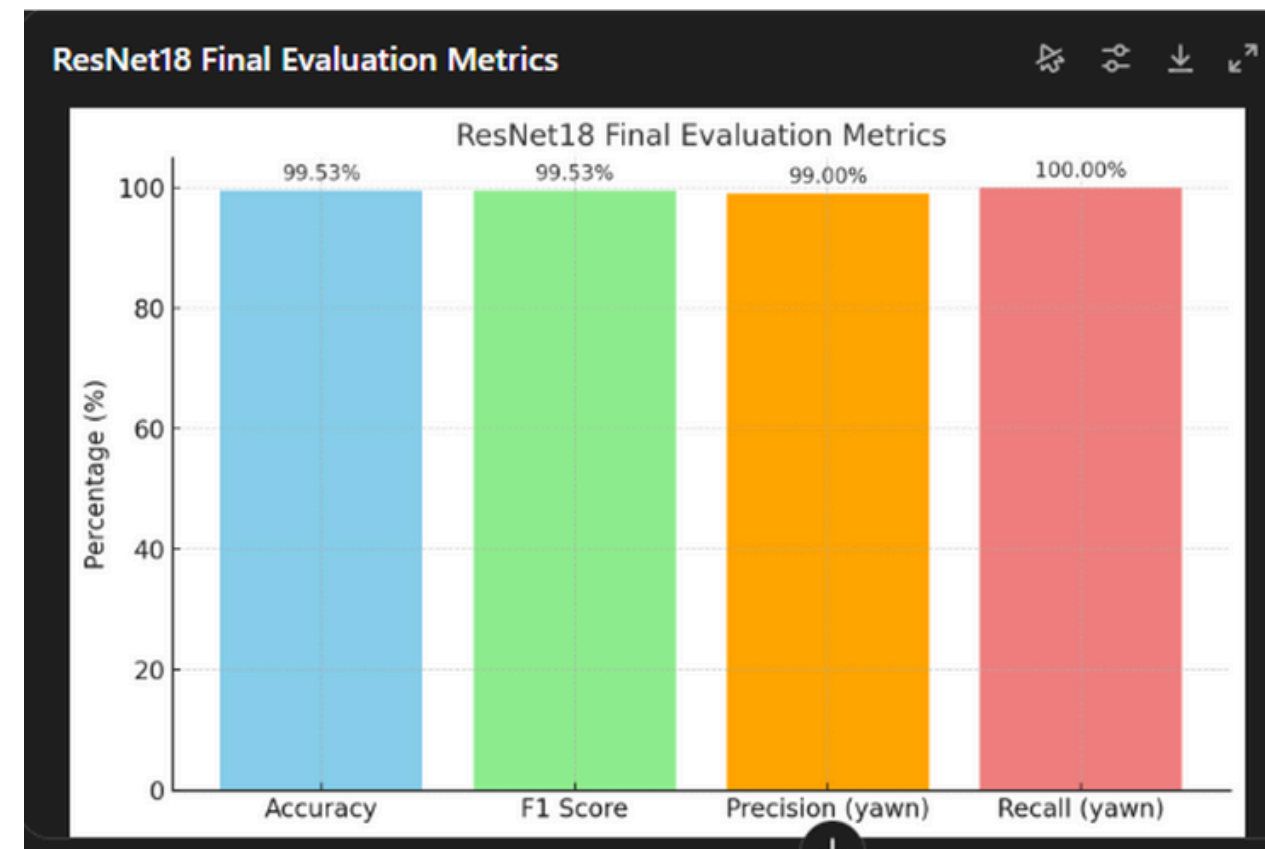
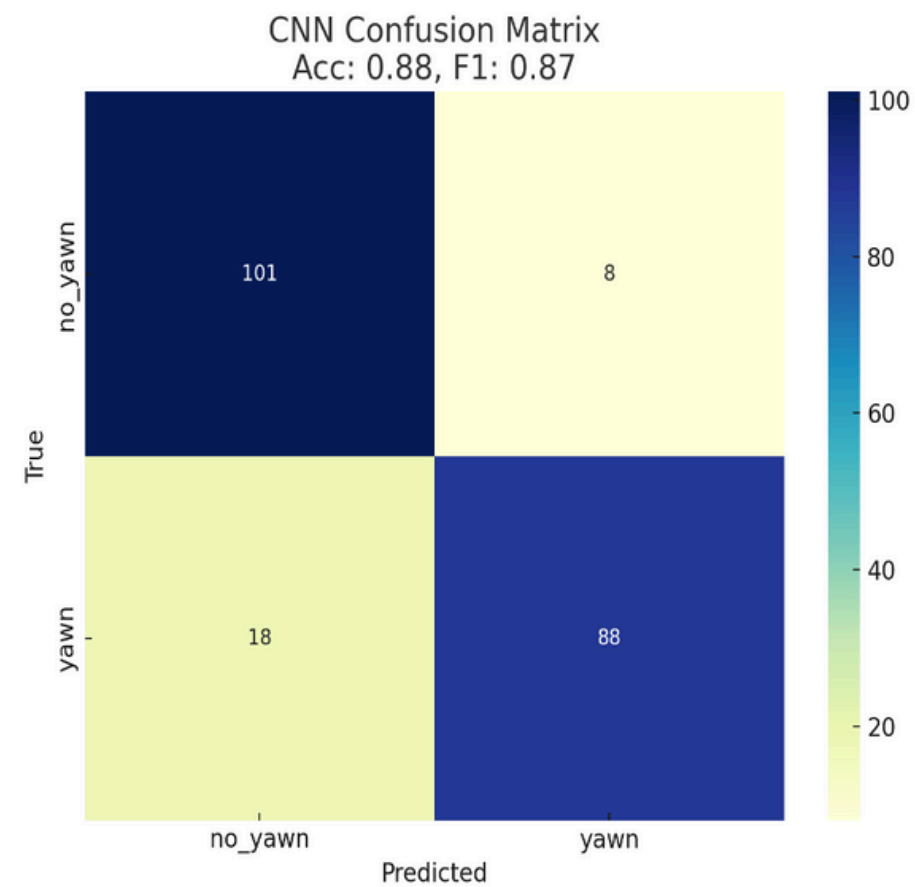


- Both models perform equally well numerically, with ~99% accuracy and F1-score
- **Grad CAM** reveals a **major difference** in how they make **predictions**
- Custom CNN consistently focuses on eye region, making it more interpretable and reliable
- ResNet18, while powerful, is less precise in attention, possibly due to general-purpose pretraining

YAWN STATE DETECTION

- We trained two models for **yawn state detection** : a custom **CNN** and **ResNet18**, using augmented data.
- **ResNet18 outperformed CNN** in terms of accuracy for eye state classification and learning.
- Conducted a analysis between the **CNN** and **ResNet18** models using **classification metrics** and **visual validation**
- We used **Grad-CAM** to visualize the specific eye regions the model focused on for its predictions.

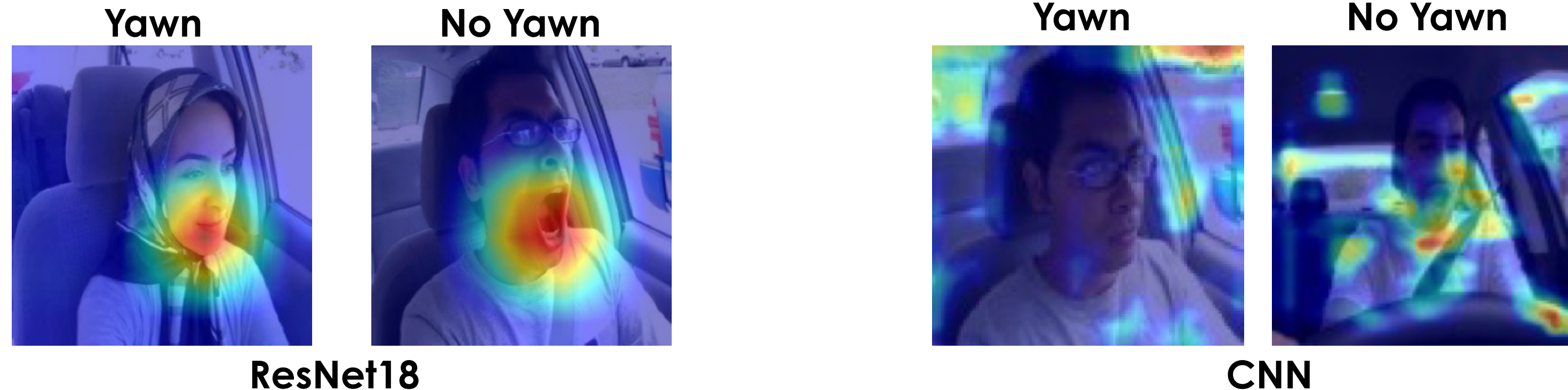
YAWN STATE DETECTION COMPARISON - CNN VS RESNET18



- ResNet18 predicted all yawning cases correctly, while the CNN made several errors.
- ResNet18 learns complex patterns (like small mouth movements), which helps it spot yawns more accurately than a basic CNN. It found every single yawn (**100% recall**) and almost never got confused (**99% precision**), making it much more reliable.

YAWN STATE (OPEN)

GRAD CAM - CNN VS RESNET18



- **ResNet18** highlights the **mouth area**, which is the key indicator for yawning, while CNN looks at random, less relevant parts.
- ResNet18's attention maps are sharp and focused, showing it **understands where to look**; CNN's maps are **scattered** and **noisy**.
- ResNet18 **ignores irrelevant areas**, whereas CNN often gets distracted, reducing reliability.

LIVE DETECTION

Eye Closure Detection

- If the driver's eyes remain closed for more than 0.5 seconds, an **"ALERT"** is triggered to indicate possible drowsiness.

Yawn Detection

- If a yawn is detected, a **"WARNING"** message appears, flagging early signs of fatigue.

LIVE DETECTION METHODOLOGY

- Webcam input video is converted into frames using OpenCV.
- **MediaPipe Face Mesh** is used to **detect eye regions** from each frame, and **MediaPipe Face Detection** is used to detect the **face region**.
- These cropped regions are then passed to the respective deep learning models for prediction.
- **Grad-CAM** overlays highlight which parts of the image influenced the model's prediction, providing transparency and trust in the detection process.

DRAWBACKS

Limited to Frontal Faces

- The current system detects eyes and yawns accurately only when the face is straight and clearly visible to the camera.
- However, at side angles or tilted head positions, detection performance drops, leading to missed alerts.

Future Improvement: Train the models on more diverse data with various head poses and angles to enhance robustness across real-world scenarios.

LIVE DEMO

Locally Hosted using streamlit
<http://localhost:8501>



THANK YOU

